ANANDRATHI

KRN Heat Exchanger Limited

25-Sep-24

SUBSCRIBE for LONG TERM

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RINKLE SHAILESH VIRA

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Issue Details

Issue Details	
Issue Size (Value in ` million, Upper Band)	3419
Fresh Issue (No. of Shares in Lakhs)	155.4
Offer for Sale (No. of Shares in Lakhs)	-
Bid/Issue opens on	25-Sep-24
Bid/Issue closes on	27-Sep-24
Face Value	Rs. 10
Price Band	209-220
Minimum Lot	65

- **Objects of the Issue**
- **Fresh Issue: ₹ 3419 million**
  - Investment in subsidiary for setting up new manufacturing facility at Neemrana Rajasthan
  - For general corporate purposes.
- **Offer for sale: NIL**

Book Running Lead Managers
Holani Consultants Pvt Ltd
Registrar to the Offer
Bigshare Services Pvt ltd

Capital Structure (` Million)	Aggregate Value
Authorized share Capital	720.0
Subscribed paid up Capital (Pre-Offer)	466.6
Paid up capital (Post - Offer)	621.5

Share Holding Pattern %	Pre Issue	Post Issue
Promoters & Promoter group	94.4%	70.8%
Public	5.6%	29.2%
Total	100.0%	100.0%

Financials

Particulars (Rs. In Million)	FY24 (Consolidated	FY23 (Standalone)	FY22 (Standalone)
Revenue from operations	**3082.8**	**2474.8**	**1561.1**
Operating expenses	2498.3	1981.6	1391.8
EBITDA	**584.5**	**493.2**	**169.3**
Other Income	52.5	24.0	21.1
Depreciation	39.9	31.7	25.3
EBIT	**597.1**	**485.5**	**165.1**
Interest	39.8	30.5	15.4
PBT	**557.3**	**455.0**	**149.7**
Tax	166.6	131.8	43.7
Consolidated PAT	**390.7**	**323.2**	**106.0**
EPS	**6.3**	**5.2**	**1.7**
Ratio	**FY24 (Consolidated**	**FY23 (Standalone)**	**FY22 (Standalone)**
EBITDAM	19.0%	19.9%	10.8%
PATM	12.7%	13.1%	6.8%
Sales growth	24.6%	58.5%	

Company description

Established in 2017 in Rajasthan KRN heat exchanger is leading manufacturer and exporter of aluminum, copper fins and copper tubes heat exchangers, water coils, condenser and evaporator coils. All the products offered are used in the domestic, commercial, and industrial Heating, Ventilation, air conditioning and refrigeration (HVAC&R) industry.

The company manufactures heat exchangers of various shapes and sizes as per the requirements of the customers demand in the market. The diameter of heat exchanger tubes it produces ranges from 5MM to 15.88 MM. Products cater to a diverse range of applications including air conditioning, heating, ventilation, refrigeration, and process cooling across various industries such as residential ACs, commercial HVAC, data center cooling, railways, automotive, defence, civil, and commercial air conditioning.

KRN served 113 customers with a mix of increasing exports moving from 10% to 15% over FY22- 24 expanding customer network across Europe and diversifying customer base and increasing market share. With exports beginning from 2019 it boasts a wide customer base across India and globally, with partnerships with industry leaders like Daikin, Schneider Electric, Kirloskar, Blue Star, and others. The company also export products overseas to countries including UAE, USA, Italy, Saudi Arabia, Norway, Czech Republic, Germany, UK, and more.

The company's manufacturing unit is equipped and capable of carrying out end-to-end manufacturing activities located at renowned RIICO industrial areas of Neemrana, Rajasthan. It has experience in precision engineering, strategically located production facilities, and focus on design, coupled with technologically advanced and cost-competitive manufacturing technology processes have enabled us to meet our customers' bespoke and stringent requirements. KRN have full-service capabilities across the product cycle including product design and development, material sourcing, designing, testing and measurement infrastructure, all under one roof to meet the requirement of our global customers.

Over the last 3 Financial Years, KRN's revenue from operations has increased at a CAGR of 25.46% out of which export sales have increased at a CAGR of 44.71%. The new plant will be commissioned by April 2025, and with its contribution, the company expected to increase its business volume by 7X in coming few years. In fact, it is the first mover in this segment to get listed within a span of just around seven years with gradual progress in its product ranges and the volume.

Valuation & outlook

KRN Heat Exchanger and Refrigeration Limited (KRN) specializes in manufacturing fin and tube type heat exchangers for the heat ventilation air conditioning and refrigeration (HVAC&R) industry and their products are primarily made from non-ferrous metals such as copper and aluminium. KRN plans to expand its capacity multi-fold which will generate increased growth in revenues from existing as well as new products like bar and plate heat exchanger, oil cooling unit with blower and motor and roll bond evaporator with improved margins from the export markets.

On Valuation parse, at the upper price band the company's implied market cap is around Rs 13100 Mn valuing at PE of 33.5 times for FY24 which appears to be fully priced. Looking at the company's experienced legacy along with its plan to expand internationally and rising demand in HVAC market which could aid KRN's revenue and profit margin in long run. Therefore we believe that this issue may be considered for its long term growth. Hence we give "**SUBCRIBE for LONG TERM"** rating to this IPO.

Company's Operations

KRN Heat Exchanger and Refrigeration Limited (KRN) specializes in manufacturing fin and tube type heat exchangers for the heat ventilation air conditioning and refrigeration (HVAC&R) industry and their products are primarily made from non-ferrous metals such as copper and aluminum. KRN customizes heat exchangers in various shapes and sizes to meet customer requirements and market demand with diameters ranging from 5 mm to 15.8 mm. The product range includes condenser coils, evaporator units, evaporator coils, header/copper parts, fluid and steam coils and sheet metal parts for various cooling applications.

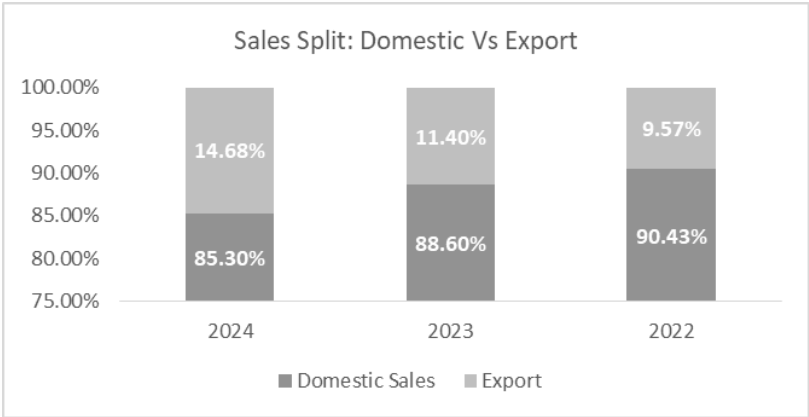
To enhance the longevity and corrosion resistance of heat exchangers and their components, various types of coatings such as nano coating, and powder coating are applied to the surfaces of copper tubes, brazing joints, and fin sheets. By applying these advanced coatings to heat exchangers and their components, KRN products ensure that the heat exchanger operates efficiently over an extended period, reducing maintenance costs and enhancing overall performance. The protective barrier provided by these coatings ensures that the heat exchanger withstands harsh conditions and maintains its optimal functionality.

Through this forward integration, the company is planning to venture into the distribution and sells channels which allow them to control the marketing and distribution of complete product range of heat exchangers. This enables them to capture a greater share of the value chain and establish direct connections with end customers. By incorporating KRN HVAC Products Private Limited, the Company can have greater control over the supply of HVAC industry products, enhancing customer reach and brand visibility.

The Company has a wide customer base covering various parts of India as well as across the world. Their prestigious customers, including Daikin Air conditioning India Private Limited, Schnieder Electric IT Business India Private Limited, Kirloskar Chillers Private Limited, Blue Star Limited, Climaventa Climate Technologies Private Limited and Frigel Intelligent Cooling Systems India Private Limited among others, attest to our commitment to delivering top-notch products and outstanding services.

	Fiscal 2024 (Rs Millions)		Fiscal 2023(Rs Millions)		Fiscal 2022 (Rs Millions)	
Particulars	Amount	% of Revenue from operations	Amount	% of Revenue from operations	Amount	% of Revenue from operations
Top 1 customer	1027.9	33.4%	813.0	32.5%	503.9	32.2%
Top 5 customer	1699.3	55.1%	1318.4	55.8%	933.4	59.7%
Top 10 customer	2229.5	72.3%	1754.0	70.8%	1183.5	75.8%

The company also export products overseas to countries including UAE, USA, Italy, Saudi Arabia, Norway, Czech Republic, Germany, UK, and more. Its manufacturing facility is situated in Neemrana, Rajasthan over an aggregated area of 7,800 sq. mt. The company uses the latest coil manufacturing machines including Hair Pin Bender, Fin Press Machine, CNC Tube Bender, and Vertical Expander.



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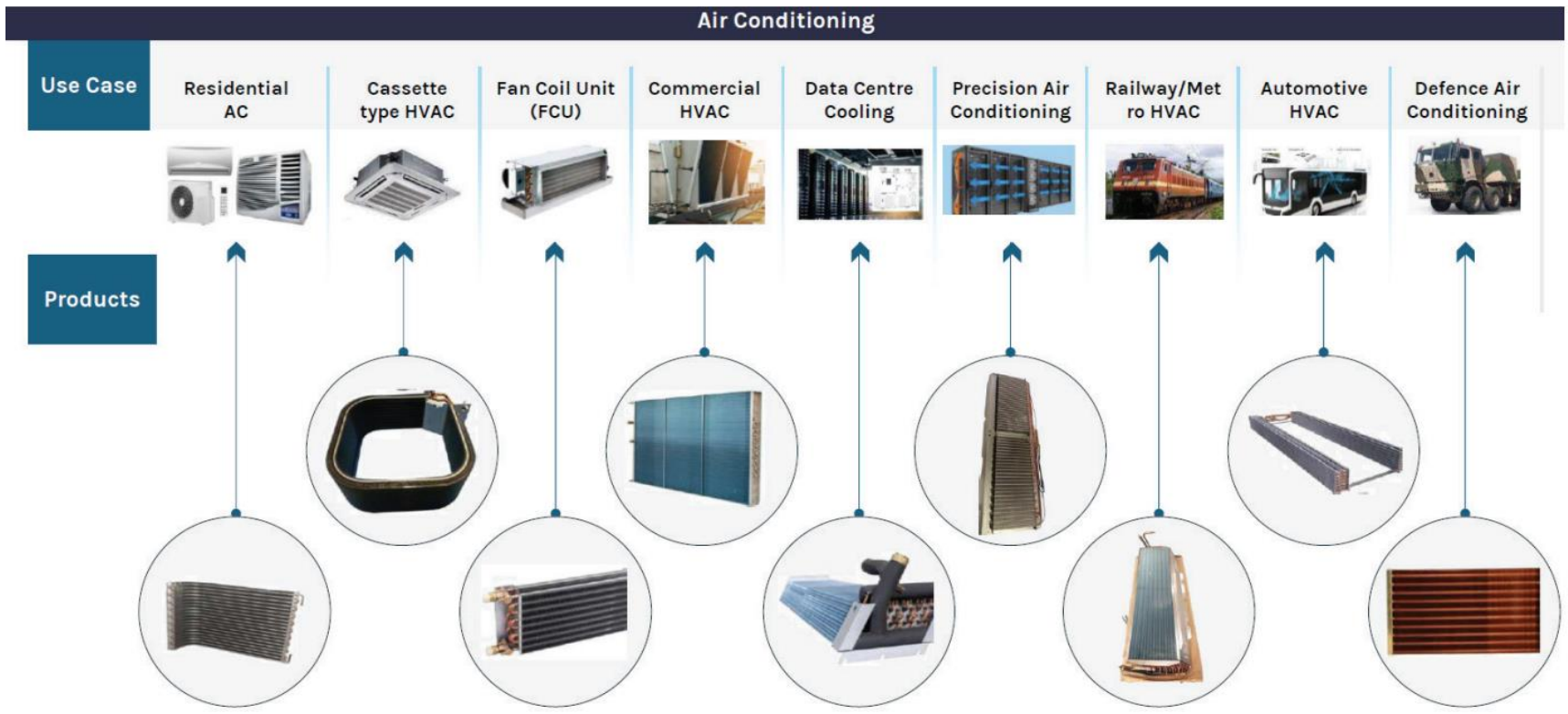
	Fiscal 2024 (Rs Millions)		Fiscal 2023(Rs Millions)		Fiscal 2022 (Rs Millions)	
Particulars	Amount	% of Revenue from operations	Amount	% of Revenue from operations	Amount	% of Revenue from operations
Evaporator Coils	1155.8	37.4%	715.3	28.8%	514.4	32.9%
Condenser Coils	1702.9	55.2%	1522.6	61.5%	994.6	63.7%
Headers/ Copper Parts	46.7	1.5%	18.1	0.7%	5.7	0.3%
Sheet Metal Parts	1.4	0.05%	1.6	0.07%	0.55	0.04%
Other Operating Revenue*	175.8	5.7%	216.9	8.7%	45.8	2.9%

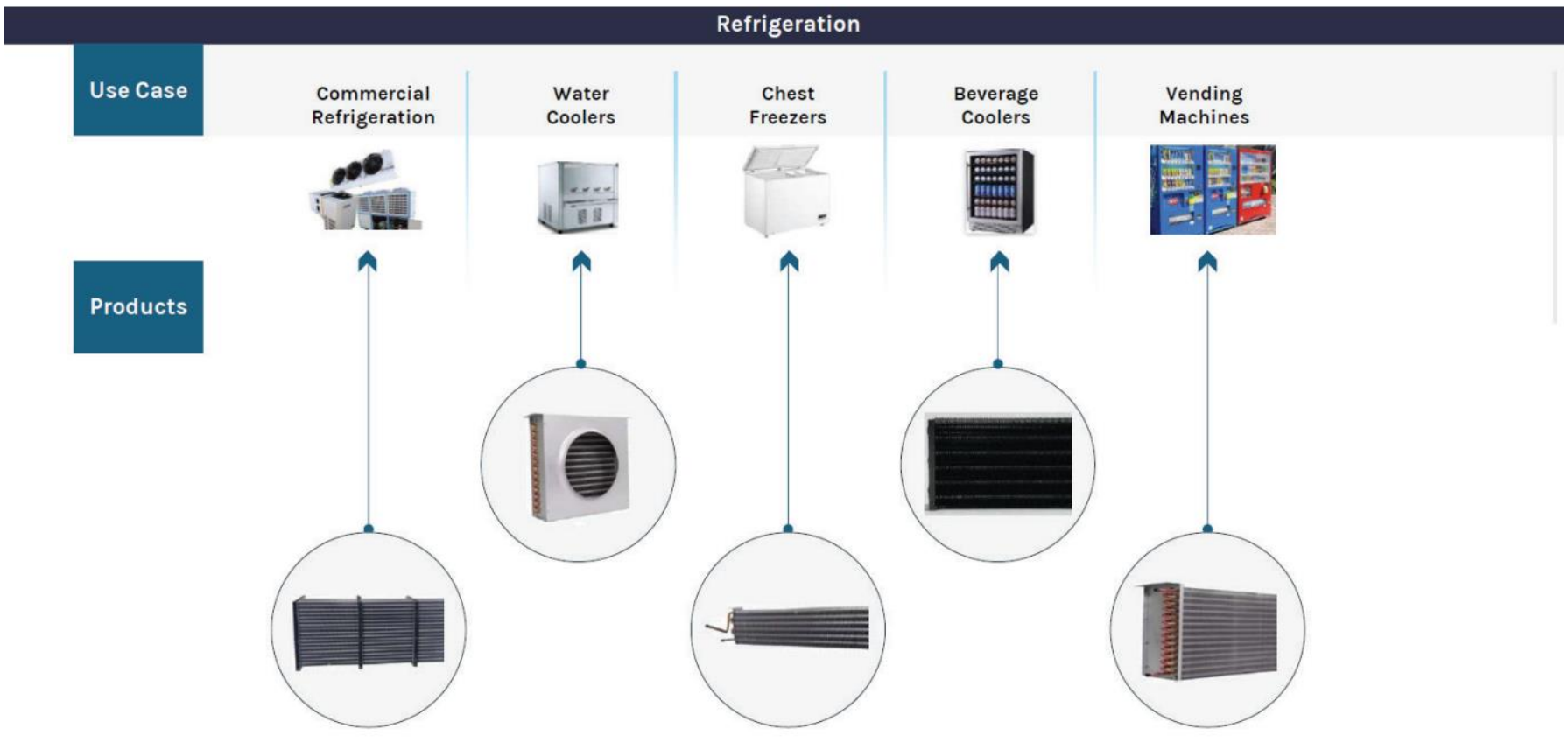
*Other Operating revenue includes sale of other HVAC parts like Tubing, Copper Pipes, rework of coils and HVAC parts etc

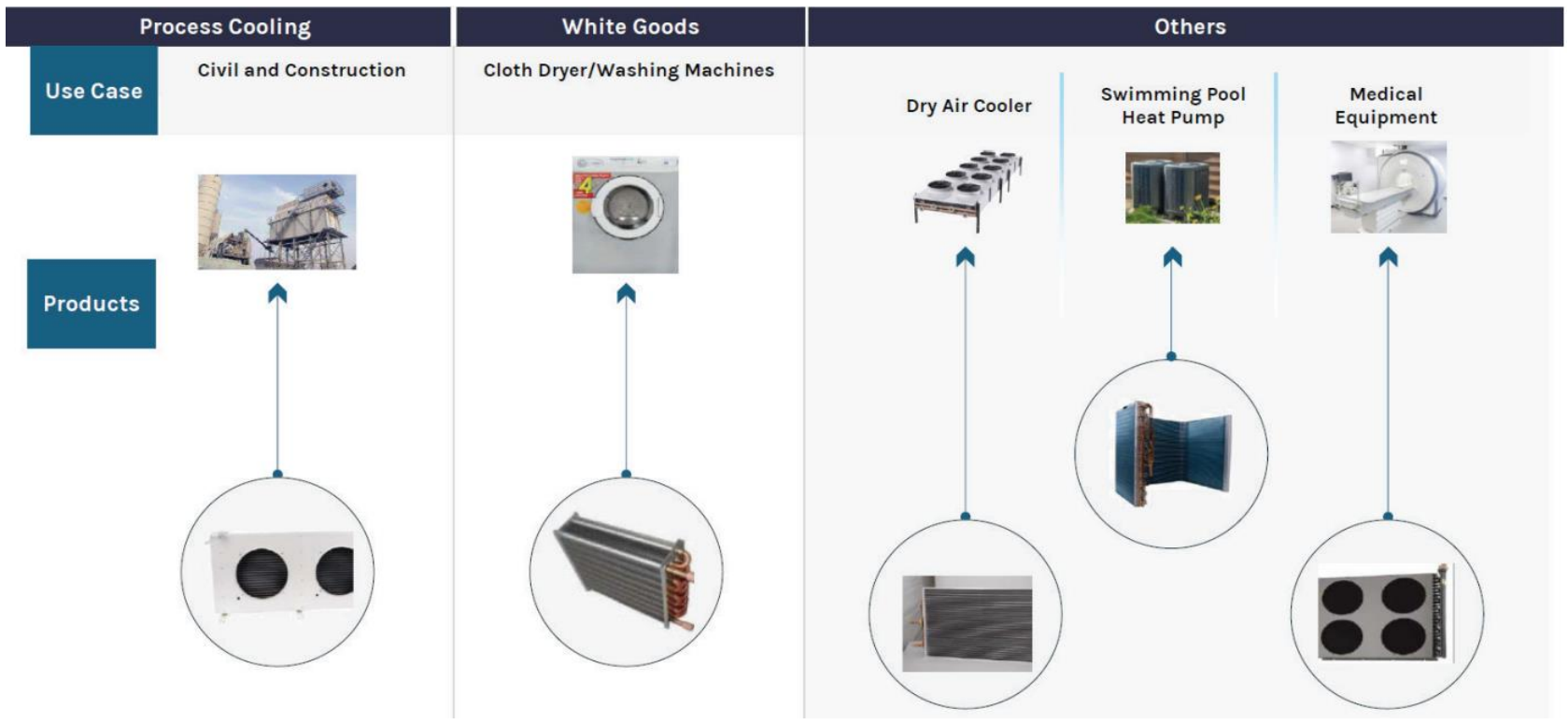
Operational KPIs of the company

Particulars	For the Financial Year ended on		
	March 31, 2024	March 31, 2023	March 31, 2022
Total Quantity Sold (in Units)			
Evaporator Coils	*2,02,273*	*1,13,182*	*1,07,564*
Condenser Coils	*2,45,950*	*2,74,890*	*2,25,063*
Headers/ Copper Parts	*1,29,580*	*96,630*	*14,977*
Sheet Metal Parts	*1,141*	*7,138*	*652*
Total Quantity Produced (In units)			
Evaporator Coils	*2,20,152*	*1,13,027*	*1,07,356*
Condenser Coils	*2,56,264*	*3,05,944*	*2,26,300*
Headers/ Copper Parts	*6,06,854*	*4,85,552*	*3,60,118*
Sheet Metal Parts	*4,77,557*	*4,22,214*	*3,34,090*
Total Number of Customers served during the year	***113***	***112***	***95***
- International Clients	*32*	*16*	*10*
- Domestic Clients	*81*	*96*	*85*
Total Number of Employees	***538***	***478***	***307***
- Permanent Employees	*147*	*124*	*92*
- Trainee	*24*	*32*	*38*
- Contractual Employees	*367*	*322*	*177*

Existing Product portfolio







New Product portfolio

Strengths:

Long-standing business relationships with leading clientele

The company focus on building sustained and long-term client relationships with their clients and constantly try to cater to customer needs with products in demand. Their experience in automated production facilities is focused on research and development, coupled with technologically advanced, quality consistency on-time delivery and cost-competitive manufacturing technology processes has resulted in repeat orders from the key customer groups. The long-term relationship with their customers allows them to understand and cater to their diverse requirements, including the development of new products and helps the company solicit new business from potential customers in the same industry.

Client's Name	Association Since
Daikin Airconditioning India Private Limited	2018
Schnieder Electric IT Business India Private Limited	2019
Kirloskar Chillers Private Limited	2020
Blue Star Limited	2021
Climaventa Climate Technologies Private Limited	2021

Established Manufacturing facility

KRN manufacturing unit is equipped and capable of carrying out end-to-end manufacturing activities located at renowned RIICO industrial areas of Neemrana, Rajasthan. It has experience in precision engineering, strategically located production facilities, and focus on design, coupled with technologically advanced and cost-competitive manufacturing technology processes have enabled them to meet their customers' bespoke and stringent requirements. KRN have full-service capabilities across the product cycle including product design and development, material sourcing, designing, testing and measurement infrastructure, all under one roof to meet the requirement of their global customers.

Effective quality control checks

Products go through stringent quality control checks such as Helium Leak Test, Water Leak Test, Contamination Test, Salt Spray Testing, Burst Pressure Testing and Pressure Cyclic Testing to ensure the best quality products are supplied to their customers. The raw materials undergo a quality check, and they have internal procedures in place for procurement of the raw materials as the quality of the final product is dependent on them. Similarly for the quality checks of their final products, company units have dedicated personnel responsible for monitoring the parameters of equipment and strength of

materials, reporting any irregularities in the manufacturing process and making adjustments accordingly. These multi-level quality checks ensure that company consistently provide good quality products, enhancing their brand value.

Promoters exhibit strong leadership and are supported by highly experienced senior management

The company management team comprised of Promoters who are a qualified engineers and have extensive experience and know-how in the engineering sector, including, business development, operations, administration, marketing, and human resource management. It leverage the understanding and experience of their management to successfully oversee our operations and growth. The founder, Promoter and Director Santosh Kumar Yadav has an experience of over 19 years in the business of manufacturing heat exchangers and refrigeration units. His responsibilities encompassed overseeing production processes, ensuring quality standards, and leading a team of engineers. It also benefit significantly from the qualified and experienced employees and workforce who possess the technical capability to further expand their business and operations.

Consistent financial performance

The Company has grown from a single product to a multi-product manufacturing Company. They have demonstrated consistent growth in terms of revenues and profitability. The recognition of their product quality has enabled them to penetrate the heat exchanger products market and cater to new customers in addition to from their existing customer network. Increased acceptance of newer product categories due to improved designs and efficient pricing has further established reputation. KRN financials have seen an increase in revenue from 156 Cr in FY22 to 308 Cr in FY24 which grew by a CAGR of 25% with strong order book, repeat orders from large customer groups as it enjoys strong relationships.

Strategies

Expanding their customer base

KRN served 113 customers with a mix of increasing exports moving from 10% to 15% over FY22- 24 expanding customer network across Europe and diversifying customer base and increasing market share. This enables us to capture a greater share of the value chain and establish direct connections with end customers enhancing customer reach and brand visibility. Rapid industrialization, urbanization, and infrastructure development have driven strong demand for heat exchangers in India driven by real estate construction, manufacturing, warehousing, and transportation sectors. The Indian HVAC market, valued at USD 7.8 billion in FY21 is projected to reach nearly USD 27.4 billion by FY30

Focus on increasing operational efficiencies to improve returns

KRN is expanding the business to reap the benefits of forward integration and incorporated a wholly owned subsidiary KRN HVAC Products Private Limited to increase value-added products in our existing portfolio, venture into the distribution and sells channels which allow us to control the marketing and distribution of complete product range of heat exchangers. KRN is expanding its capacity multifold which will generate increased growth in revenues from existing as well as new products like bar and plate heat exchanger, oil cooling unit with blower and motor and roll bond evaporator with improved margins from the export markets.

Enhancing existing products base and product quality

Offering high quality and safety complying products is of much importance for customer retention and repeat-order flow. The company intend to have close interaction with their customers to strengthen their relationships with them and enable them to understand the market perception and demand for their products. Similarly, to increase their existing portfolio, it has incorporated their wholly owned subsidiary namely KRN HVAC Products Private Limited on April 07, 2023 wherein company is setting up a new manufacturing facility to manufacture products that are at present not manufactured by them, using the Net Proceeds of the Issue.

S. No.	Products	Application	Use Case
1	Bar & Plate Heat Exchanger	The Bar & plate Heat Exchanger is a device that permits recovery of the Heat contained in a fluid being transferred to another fluid. Both fluids never touch each other because they are separated by metallic sheets. These sheets, which are called plates, are very Fine and grooved to enable the diffusion of the greatest amount of Heat through each surface unit. The plate Heat Exchanger has been made to guarantee a Heat interchange with the highest security.	1. It is used in Railway locomotive for transformer and convertor oil cooler. 2. It is used in screw & centrifugal air compressors as air cooler. 3. Used as Metro train engine cooling unit and Transmission oil cooling units. 4. It is used in defence vehicles (Tanks) as radiator and oil cooler. 5. Used as charge air coolers and oil coolers of Diesel Engine or DG sets. 6. Used as oil cooler for heavy earth moving vehicles. 7. It is used in Oil rigs and mining equipment's.
2	Oil Cooling Unit with Blower & Motor	An oil cooler is a separate, smaller radiator to an engine's main radiator, which maintains an oil supply at a consistent, optimal temperature. It's purpose is to cool the oil passing through the coils, thus improving the engine and the transmission's lifetime. They are situated in front of an engine's cooling system.	1. It is specifically used as a cooling unit for locomotives/Railways. 2. It is also used in automotive vehicles as cooling unit.
3	Roll Bond Evaporator	Roll-Bond evaporators provide unique design flexibility for direct cooling refrigeration systems. Roll-Bond evaporators deliver efficient thermal performance in a product that can be shaped to fit most applications.	It is used in refrigerators as an evaporator/freezer

Increasing global footprint and augmenting growth in current geographies

With a view to further diversify their customer base and increase the market share, KRN intend to augment sales in the markets where they sell their products as well as expand into new geographies. Over the last several years, the company has consistently expanded its customer network across Europe, and this continues to be one of the core strategies of the company for the future. The company will continue to focus their efforts on selected geographies such as North America and Europe and establish a greater presence in these regions. Their growth strategy in these markets will be to create a strong local presence and foster connections, and leverage expertise with required development capabilities to exploit the growth potential offered by these markets. The company strong focus will remain on acquiring new customers, retaining existing customers and offering good quality products.

Industry Snapshot

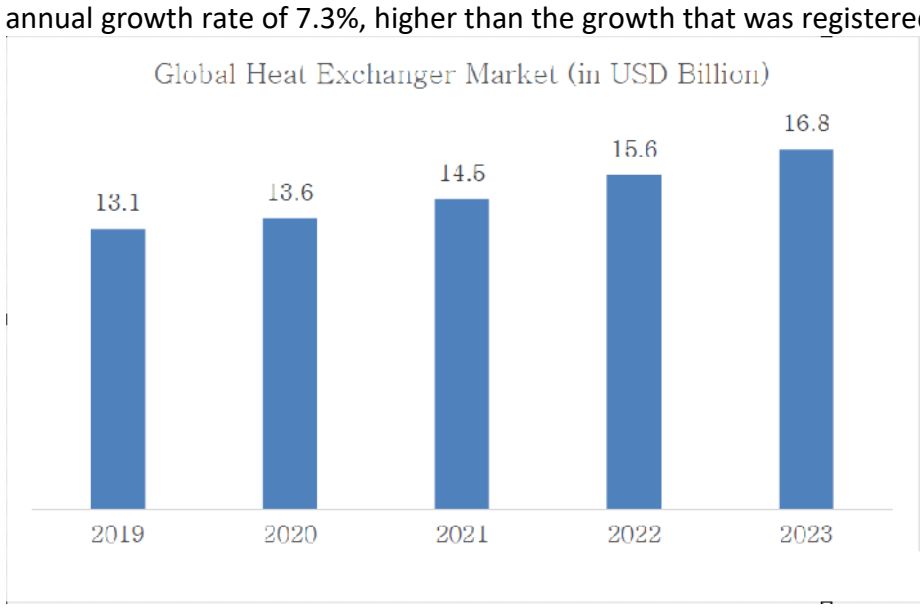
Product Profile: HEAT EXCHANGER

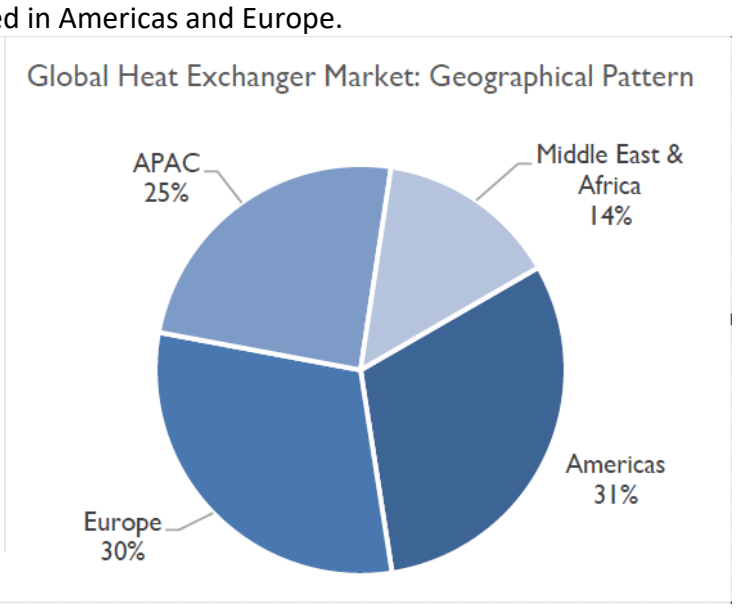
Heat exchangers transfer thermal energy from one fluid to another, while ensuring that the fluids do not mix. This process is facilitated by conduction, enabling the heat exchanger to heat or cool a substance. Heat exchangers provide a means of regulating temperature, enhancing efficiency, averting overheating, and mitigating other potential hazards, resulting in an improvement in safety standards. Heat exchangers are considered more efficient and reliable in operations wherein some parameters, like temperature differential, flow rate and installation guidelines, are evaluated successfully. Being an efficient source of energy saving, heat exchangers find applications across a broad range of domestic and industrial applications such as steam power plants, chemical processing plants, food processing, HVAC systems, mining, pulp & paper, metallurgy etc. Heat exchangers typically are suitable for deployment in areas which have ample spaces such as industrial complexes, commercial space, hospitals, educational institutes etc. These systems are not generally preferred or feasible for smaller capacities, especially residential, as the economies of scale are difficult to overcome.

Type of Heat Exchanger	Common materials used
Shell & Tube	Carbon steel, stainless steel, brass, and titanium
Finned Tube/ Air cooled	Duplex stainless steel, carbon steel, aluminum, copper
Plate Heat Exchanger	Stainless steel, Titanium

Global Heat Exchanger Market

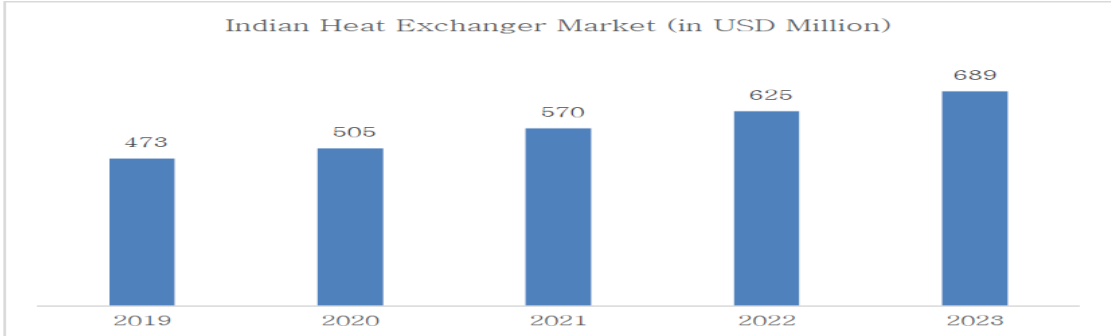
The global heat exchanger market is valued at USD 16.8 billion in 20234, with the industry experiencing a robust CAGR of 6.4% from 2019 to 2023. This growth is primarily driven by the increasing global demand for sustainable, low-energy consumption solutions that are also cost-effective. This shift has spurred significant innovation in both the products and systems used within the industry. For heat exchangers, this innovation is focused on developing superior designs that enhance thermal efficiency, thereby reducing energy costs and minimizing carbon footprints. The push towards more efficient and environmentally friendly solutions is shaping the future of the global heat exchanger market, aligning with broader goals of sustainability and technological advancement. The robust manufacturing base in these countries, coupled with expanding industrial sectors and significant public and private investments, has created a favorable market environment for heat exchangers in the APAC region, reshaping the global market dynamics. This transformation has made APAC the fastest growing heat exchanger market. Between 2019 and 2023, the APAC heat exchanger market witnessed a compounded annual growth rate of 7.3%, higher than the growth that was registered in Americas and Europe.





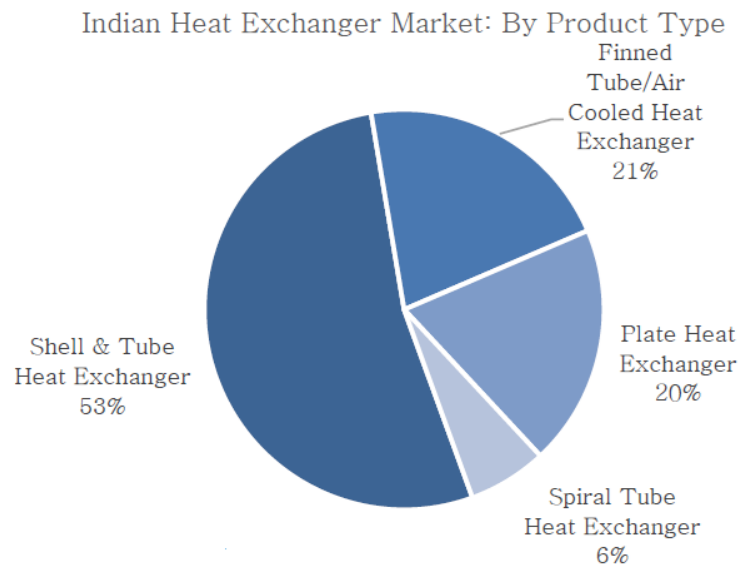
Indian Heat Exchanger Industry

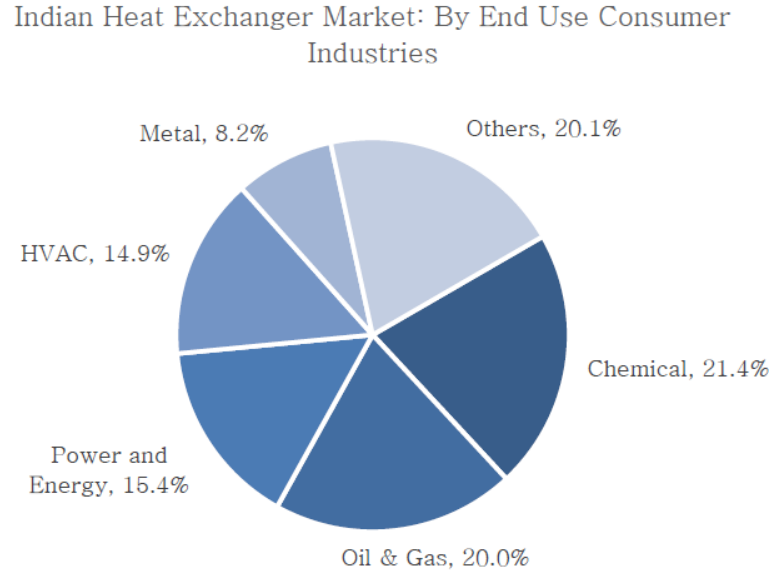
The Indian market for heat exchanges reached USD 689 million per annum in 2023, the annual industry turnover increasing by a CAGR of 10% between 2019 and 2023. The heat exchanger industry too was impacted by the spread of the Covid-19 pandemic, as annual revenue growth in the industry slowed to 6% in 2020. Demand has recovered in the subsequent years, helping the industry grow its revenue by an average of 11% per annum in 2021, 2022 and 2023.



By Type of Heat Exchanger and By End Use Industries

Shell & tube heat exchanger, followed by finned tube heat exchanger is two of the largest product segments in Indian heat exchanger industry in 2023. Shell & tube heat exchangers accounted for nearly 53% of the total industry revenue while finned tube heat exchangers accounted for nearly 21%. Between 2020 and 2023, revenue growth in shell & tube exchangers and finned tube heat exchangers has grown by a CAGR of approximately 11.4% and 10.7%. Chemical manufacturing and oil & gas industries are the two largest consumers of heat exchangers in India, with each accounting for nearly one-fifth of total industry turnover in 2023. Other key end-use consumers are power & energy, HVAC, and metallurgy. Together, these five industries contributed to 80% of the heat exchanger industry revenue in 2023.





Demand from HVAC Segment

In HVAC segment, heat exchangers are used for heating, cooling, heat recovery, dehumidification, and moisture control. The focus on creating energy efficient buildings and systems has magnified the importance of heat exchangers in a HVAC system, as deploying of heat exchangers is one of the preferred techniques to reduce energy usage as well as cut back on greenhouse gas emissions. Thus, heat exchanger finds application across HVAC industry, from residential HVAC systems to large scale industrial HVAC systems.

Emerging Trends in Indian HVAC Industry & Impact on Heat Exchangers

The HVAC market in India is on a strong growth trajectory, fueled by rapid urbanization, rising disposable incomes, and the increasing need for energy-efficient solutions amidst changing climatic conditions. The shift towards smart HVAC systems is becoming a key trend, with consumers prioritizing intelligent solutions that offer enhanced controls, real-time monitoring, and IoT integration. This is further amplified by technological advancements, such as variable refrigerant flow (VRF) systems and predictive maintenance driven by machine learning, which are enhancing energy efficiency, reducing operational costs, and contributing to sustainability in both residential and commercial segments.

Government initiatives like the 'Make in India' campaign, Atmanirbhar Bharat, and Production Linked Incentive (PLI) schemes are bolstering local manufacturing, reducing import dependency, and supporting the nation's goal of achieving carbon neutrality by 2070. Moreover, the significant government investment of USD1.45 trillion in infrastructure over the next five years is expected to drive unprecedented growth in the sector. The projected increase in the number of Indian cities with populations exceeding one million by 2030, alongside the anticipated construction boom, underscores the vast potential for the HVAC market. The growing emphasis on green building certifications like LEED and GRIHA further creates opportunities for the adoption of energy-efficient HVAC systems. Additionally, the rise of aftermarket services, including maintenance, retrofitting, and upgrades, presents new avenues for HVAC service providers and contractors, making the Indian HVAC market a compelling landscape for both local and international players.

No Comparison with Listed Peers:

Key Risks

- **Project Implementation Risks**: It plans to invest ₹22,688.54 lakhs in a new manufacturing facility. However, delays or cost overruns in machinery acquisition and construction could significantly exceed the budgeted ₹24,410.98 lakhs, impacting financial stability.
- **Customer Concentration Risk**: The Company derives substantial revenue from a few key customers, with 75.94% from the top 10 in 2023. Dependence on a few key customers for significant revenue poses the risk of revenue loss if these customers reduce orders or face financial difficulties.
- **Import Dependency and Restrictions**: 80.58% of the company's raw materials were imported as of September 2023. High reliance on imported raw materials and potential restrictions on imports by governmental authorities may disrupt operations and affect financial performance.
- **Foreign Exchange and Interest Rate Risks**: Exposure to fluctuations in foreign exchange rates and interest rates may adversely affect financial performance.
- **Geographical Concentration Risk:** Concentration of manufacturing facilities in one geographic location exposes the business to adverse developments affecting that region.
- **Transportation Risks**: Dependence on third-party transportation providers and potential disruptions in transportation services pose risks to supply chain efficiency and business operations.
- **Raw Material Price Fluctuations:** Exposure to fluctuations in raw material prices, particularly aluminum and copper, without long-term supply agreements, impacts production costs and profitability.

Valuation & Outlook

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DISCLAIMER:

Analyst Certification

- The views expressed in this Research Report accurately reflect the personal views of the analyst(s) about the subject securities or issuers and no part of the compensation of the research analyst(s) was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the research analyst(s) in this report. The research analysts are bound by stringent internal regulations and also legal and statutory requirements of the Securities and Exchange Board of India (hereinafter "SEBI") and the analysts' compensation are completely delinked from all the other companies and/or entities of Anand Rathi, and have no bearing whatsoever on any recommendation that they have given in the Research Report.

Ratings Methodology

- Analysts' ratings and the corresponding expected returns take into account our definitions of Large Caps, Mid-Caps & Small Caps as described in the Ratings Table below:

	Buy	Hold	Sell
Large Caps (Top 100 companies)	**>15%**	**0%-15%**	**Below 0%**
Mid-Caps (101st-250th company)	>20%	0%-20%	Below 0%
Small Caps (251st company onwards)	>25%	0%-25%	Below 0%

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